

TREATMENT AND EMERGENCY RECOMMENDATIONS FOR 3-Hydroxy-3-Methylglutaryl CoA Lyase Deficiency (HMG-CoA Lyase Deficiency)

RE:

DOB:

Please see last section of this letter for Emergency Treatment during Decompensation

Definition: HMG-CoA Lyase deficiency is a disorder of fatty acid metabolism and leucine metabolism. Complications can arise with illness and physiologic stress leading to hypoketotic hypoglycemia, metabolic acidosis, liver disease, and elevated ammonia.

Signs of Decompensation: Vomiting, lethargy, decreased activity and responsiveness, metabolic acidosis with low blood sugar and low or absent urine ketones, coma, death.

Treatments:

- A. ***Prevention of decompensation:***
 1. Therapeutic Diet: low fat, limited leucine (low protein), and supplementation with L-Carnitine.
- B. ***Treatments during an illness*** (colds, flu, ear infections, etc):
 1. Encourage fluids containing carbohydrate:
 - i. Infant formula
 - ii. Fruit juices
 - iii. Gatorade or ginger ale
 2. Observe the child for general activity and alertness. Watch urine output for hydration status.
 3. If unable to eat or drink for more than 4 hours or if not urinating bring to the EMERGENCY ROOM.
- C. ***Treatment during potential metabolic decompensation***
(Lethargy, vomiting, decreased state of consciousness, low blood sugar, coma. Please note that not all of these symptoms need to be present):

1. IV fluids to keep the patient in a high normoglycemic state. D10 in a $\frac{1}{2}$ normal saline at greater than maintenance to keep blood glucose >100mg/dL.

Please see next page...

2. Use IV L-Carnitine 100mg/kg loading dose IV over 1hour) compatible with D10). Then use L-Carnitine 25mg/kg every 6 hours.
3. For persistent metabolic acidosis use sodium bicarbonate 1 mEq/kg IV over 60 minutes.
4. Laboratory studies: BMP, Ammonia, Urine Ketones
5. Treat the underlying and precipitating illness (infection, etc), and obtain appropriate additional laboratory studies to establish the cause of the physiological challenge.

6. **Notify the metabolic specialist immediately (after business hours this number allows you to speak with the geneticist on call:**

Kentucky Childrens Hospital

Division of Genetics and Metabolism

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(After Hours: Please call 1-800-888-5533)